for use between devices on the same constrained network, between devices and general nodes on the Internet, and between devices on different constrained networks—both joined on the Internet. This protocol is especially designed for IoT systems based on HTTP protocols. CoAP makes use of the UDP protocol for lightweight implementation. It also makes use of RESTful architecture, which is very similar to the HTTP protocol. It is used within mobiles and social-network-based applications and eliminates ambiguity by using the HTTP get, post, put and delete methods. Apart from communicating IoT data, CoAP allows secure exchange of messages by using datagram transport layer security (DTLS) protocol.

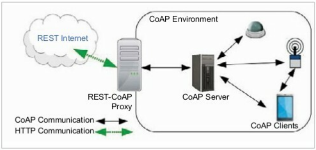

Fig. 2: How CoAP works

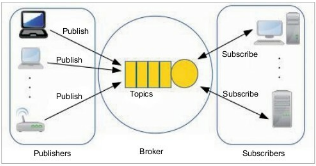

Fig. 3: MQTT protocol architecture

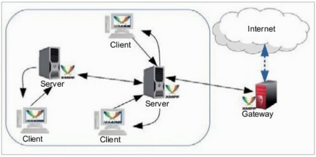

Fig. 4: XMPP architecture

MQTT protocol. Message queue telemetry transport (MQTT), a messaging protocol, was developed by Andy Stanford-Clark of IBM and Arlen Nipper of Arcom in 1999. It is mostly used for remote monitoring in the IoT. Its primary task is to acquire data from many devices and transport it to the IT infrastructure. MQTT connects devices and networks with applications and middleware. A hub-and-spoke architecture is natural for MQTT. All the devices connect to data concentrator servers like IBM's new MessageSight appliance. MQTT protocols work on top of TCP to provide simple and reliable data streams.

MQTT protocol consists of three main components: subscriber, publisher and broker. The publisher generates the data and transmits the information to subscribers through the broker. The broker ensures security by cross-checking the authorisation of publishers and subscribers.

MQTT protocol is the preferred option for IoT-based devices, and is able to provide efficient information-routing functions to small, cheap, low-memory and power-consuming devices in vulnerable and low-bandwidth networks.

Extensible Messaging and Presence Protocol (XMPP). It is a communication IoT protocol for message-oriented middleware based on XML language. It enables real-time exchange of structured yet extensible data between any two or more network entities. The protocol was developed by the Jabber open source community in 1999, basically for real-time messaging, presence information and maintenance of contact lists. XMPP enables messaging applications to attain authentication, access control, and hop-by-hop and end-to-end encryption. Being a secure protocol, it sits on top of core IoT protocols and connects the client to the server via a stream of XML stanzas. The XML stanza has three main components: message, presence and IQ.

Advanced Message Queuing Protocol (AMQP). Developed by John O'Hara at JPMorgan Chase in London, AMQP is an application-layer protocol for message-oriented middleware environments. It supports reliable communication via message delivery assurance primitives like at-most once, atleast once and exactly once delivery.

The AMQP protocol consists of a set of components that route and store messages within a broker service, with a set of rules for wiring the components together. It enables client applications to talk to the broker and interact with the AMQP model. This model has the following three components:

1. *Exchange.* Receives messages from publisher-based applications and routes them to'message queues'

2. *Message queue.* Stores messages until these can be safely processed by the consuming client application

3. *Binding.* States the relationship between the message queue and the exchange

These components are connected into processing chains in the server to create the desired functionality.

Data Distribution Service (DDS). This IoT protocol for real-time machine-to-machine communication was developed by the Object Management Group (OMG). It enables scalable, real-time, dependable, high-performance and interoperable data exchange via publish-subscribe methodology. Compared to MQTT and CoAP IoT protocols, DDS makes use of broker-